



*T(4th Sm.)-Computer Sc.-H/(SEC-B-1)/CBCS*

**2021**

**COMPUTER SCIENCE — HONOURS**

**Paper : SEC-B-1**

**(Information Security)**

**Full Marks : 80**

*The figures in the margin indicate full marks.*

*Candidates are required to give their answers in their own words  
as far as practicable.*

**Group - A**

1. Answer **any five** questions :

2×5

- (a) What is watermarking?
- (b) What is proxy firewall?
- (c) Find the value of  $3^{201} \bmod 11$ .
- (d) List out the services provided by PGP.
- (e) Find the value of  $321 \bmod 11$  using Fermat's theorem.
- (f) What is steganography?
- (g) What is logic bomb?
- (h) What is 'worm'?

**Group - B**

Answer **any four** questions.

5×4

- 2. What are the differences between block cipher and stream cipher?
- 3. Give example each for substitution and transposition ciphers.
- 4. Explain Elgamal crypto system.
- 5. Explain why SHA is more secure than MD5?
- 6. What is the purpose of X-boxes in DES?
- 7. Explain Kerberos.

**Please Turn Over**

**Group - C**

Answer **any five** questions.

8. (a) Discuss the various steps used in verification of digital certificate.  
(b) What do you mean by cryptanalysis and cryptography? 5+5
9. (a) Explain Diffie–Hellman key exchange algorithm.  
(b) What are symmetric cipher and asymmetric cipher? 5+5
10. (a) What is IP sniffing and IP spoofing?  
(b) What are the four main stages in AES operation? 5+5
11. (a) What are the limitations of firewall?  
(b) What is Brute-force-attack? Explain. 5+5
12. (a) What are the features of ITAA?  
(b) State and prove Fermat's little theorem. 5+5
13. Write short notes on (**any two**) : 5×2
- (a) S/MIME
  - (b) Firewall
  - (c) Digital Signature
  - (d) Secure Hash Algorithm (SHA)
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